

Simple2DSeg - Desktop Help Guide

Image-segmentation workflow | Windows/Linux/Headless Server interfaces

Purpose. Simple2DSeg is a Python 2D mask-creation tool. Its workflow is designed to mimic selected behavior of the MATLAB 2018 Image Segmenter app using python equivalents. The application was created with the aid of OpenAI. This guide covers the desktop Tkinter interface shown below.

Quick workflow: 1) Open the CSV and let the app load the next unprocessed image. 2) Create or load a mask. 3) Add/refine regions using the ribbon or guided tabs. 4) Review History and the display modes. 5) Choose Save & Next [1] to save and continue.

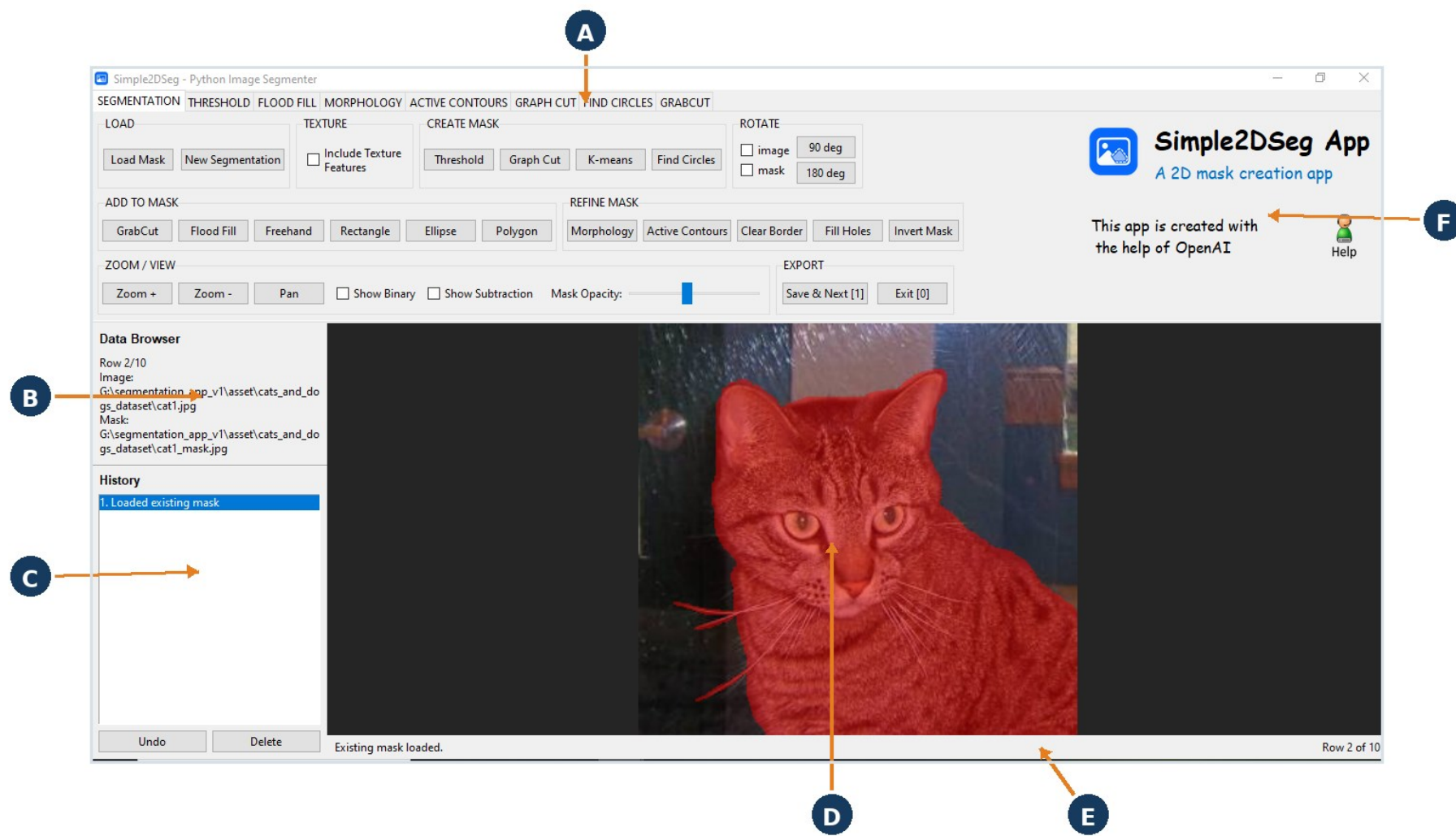


Figure 1. Simple2DSeg workspace overview. Lettered callouts are explained below.

A Tabs & ribbon | B Data Browser | C History | D Image workspace | E Status/progress | F Help & branding

Main Segmentation Ribbon

Use the numbered image as a quick map of the controls you will use most often.

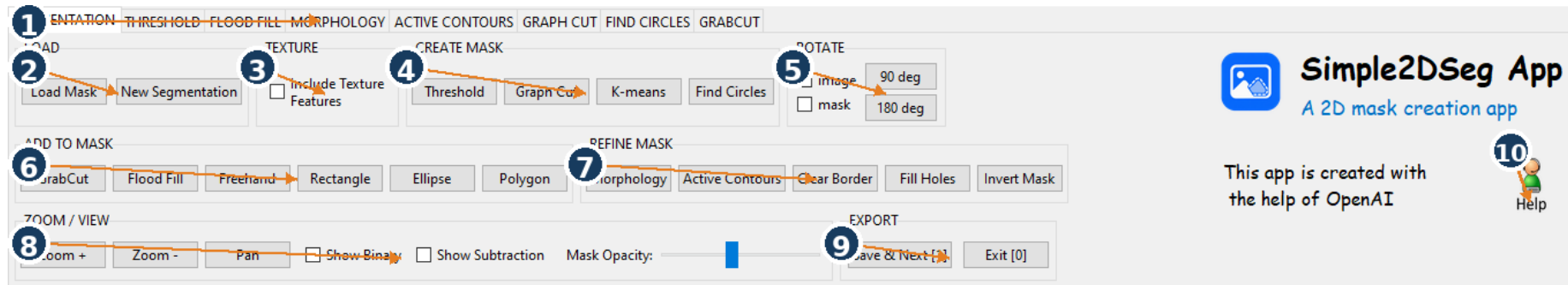


Figure 2. Main SEGMENTATION ribbon. Numbers match the reference table below.

No.	Area / control	What it does	How to use / notes
1	Tabs	Choose the main workspace or a guided segmentation method.	Most guided tabs use Preview -> Apply -> Close.
2	LOAD	Load an existing binary mask or start an empty mask.	Loaded mask dimensions must match the original image.
3	TEXTURE	Add texture descriptors to supported segmentation methods.	Used by K-means, Flood Fill, Active Contours, and Graph Cut.
4	CREATE MASK	Threshold by brightness; Graph Cut uses scribbles; K-means clusters pixels; Find Circles detects circular objects.	K-means commits immediately; the guided methods can be previewed first.
5	ROTATE	Rotate selected image and/or mask clockwise by 90 or 180 degrees for editing.	Saved masks return to original orientation; rotate both image and mask for 90 degrees.
6	ADD TO MASK	GrabCut, Flood Fill, Freehand, Rectangle, Ellipse, or Polygon add foreground regions.	For Polygon, click vertices and double-click to finish.
7	REFINE MASK	Morphology, Active Contours, Clear Border, Fill Holes, and Invert Mask clean or reshape the mask.	Active Contours needs a non-empty starting mask.
8	ZOOM / VIEW	Zoom, Pan, Show Binary, Show Subtraction, and Mask Opacity control only the display.	Binary and Subtraction views are mutually exclusive; mouse wheel also zooms.
9	EXPORT	Save & Next [1] saves outputs, updates the CSV, and advances. Exit [0] closes without saving the current row.	Keyboard: 1 = Save & Next; 0 = Exit current row without saving.
10	Help	Save a local copy of help_segment_app.pdf.	Available from every tool tab.

Guided Segmentation Tool Tabs

These tabs expose extra settings for assisted or automatic segmentation methods.

Preview / Apply / Close: Preview shows a temporary result. Apply commits it to the mask and adds a History entry. Close removes an unapplied preview and returns to the main SEGMENTATION tab.

Tool tab	Main settings	Layperson workflow
THRESHOLD	Method: Global, Manual, Adaptive; Threshold; Sensitivity; Foreground Polarity	Global chooses an automatic cutoff. Manual uses the entered cutoff. Adaptive uses local brightness with sensitivity and bright/dark polarity. Preview, then Apply if correct.
FLOOD FILL	Distance Metric: Euclidean or Geodesic; Tolerance; Select Seed	Choose settings, click Select Seed, then click inside the object. The connected region becomes a preview that can be applied to the current mask.
MORPHOLOGY	Operation: dilate, erode, open, close; Shape: disk, diamond, line, octagon, square, rectangle; size/angle controls	Use to grow, shrink, smooth, separate, or close small gaps in the current mask. Preview the effect before applying.
ACTIVE CONTOURS	Method: Chan-Vese or edge; Iterations	Refines a non-empty starting mask by evolving its boundary toward image features. More iterations allow more evolution.
GRAPH CUT	Foreground / Background / Erase scribbles; Marker Size; Clear controls; Show Boundaries; Superpixel Density	Mark a few definite foreground and background areas, then Preview. Increase density for finer superpixels when needed.
FIND CIRCLES	Min Diameter; Max Diameter; Object Polarity; Sensitivity	Searches for circular objects within the size range. Choose bright or dark objects and adjust sensitivity, then Preview.
GRAB CUT	ROI: Rectangle or Polygon; Draw ROI; Foreground / Background / Erase / Clear scribbles	Draw an ROI around the object, optionally add foreground/background corrections, then Preview and Apply. GrabCut adds its result to the existing mask.

Saving, resume behavior, and practical tips

CSV and saving behavior

- The CSV requires File_path. If Segment or Mask_path is missing, the app adds the column automatically.
- Blank Segment values are treated as unprocessed; already flagged rows are skipped.
- If Mask_path is blank, the default mask is <image>_mask.<ext>. The masked image is saved as <image>_sub.<ext>.
- Save & Next writes the binary mask, writes image x mask, stores the absolute mask path, sets Segment = 1, and loads the next row.
- Exit leaves the current row unflagged so it can be resumed later.

Useful operating notes

- Supported 2D image formats: PNG, JPG/JPEG, TIFF/TIF, and BMP.
- Preview is temporary. Apply adds the preview to History. Close discards an unapplied preview and returns to SEGMENTATION.
- History items are complete cumulative mask states. Deleting an older edit also removes later dependent states so the selected mistake is fully undone.
- Graph Cut / GrabCut display foreground marks in green, background marks in red, and ROI or superpixel boundaries in yellow.
- Rotation changes the editing view only; the original image is not overwritten, and the saved mask is returned to the original orientation.
- Mouse wheel = zoom. Polygon tools = click vertices, then double-click to finish.

Attribution note: Simple2DSeg is an independent Python implementation inspired by the MATLAB 2018 Image Segmenter workflow. MATLAB is a product of MathWorks; this tool is not a MathWorks product. The application was created with the aid of OpenAI.